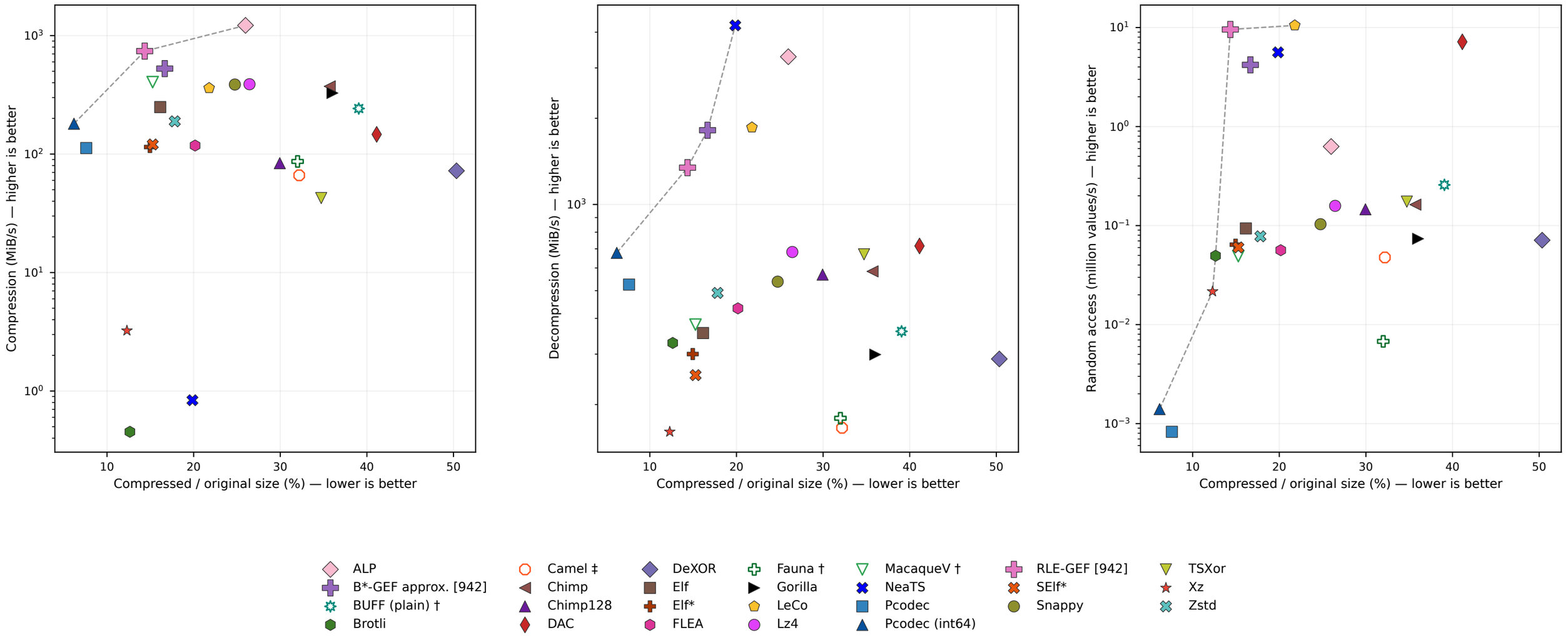
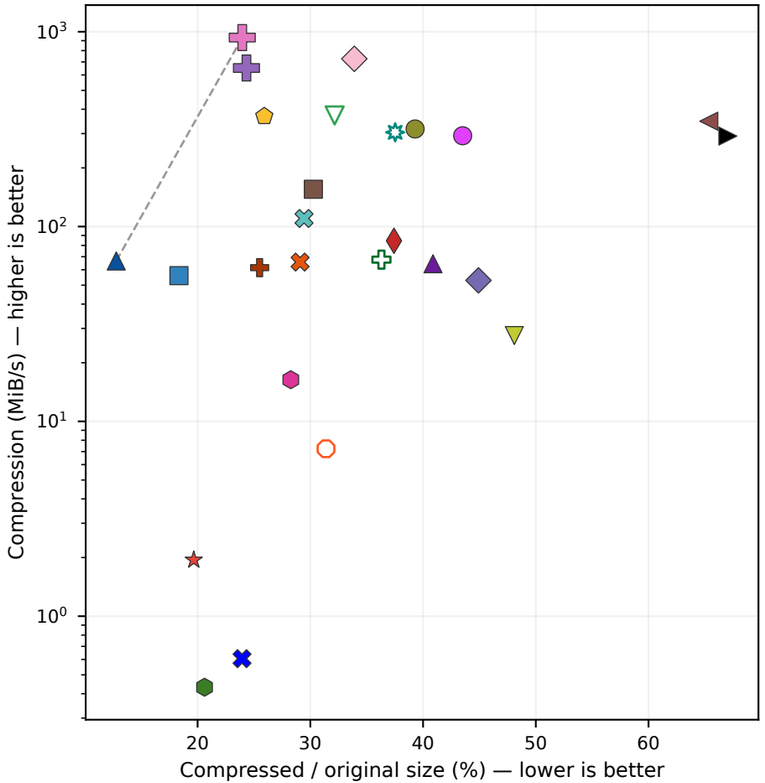
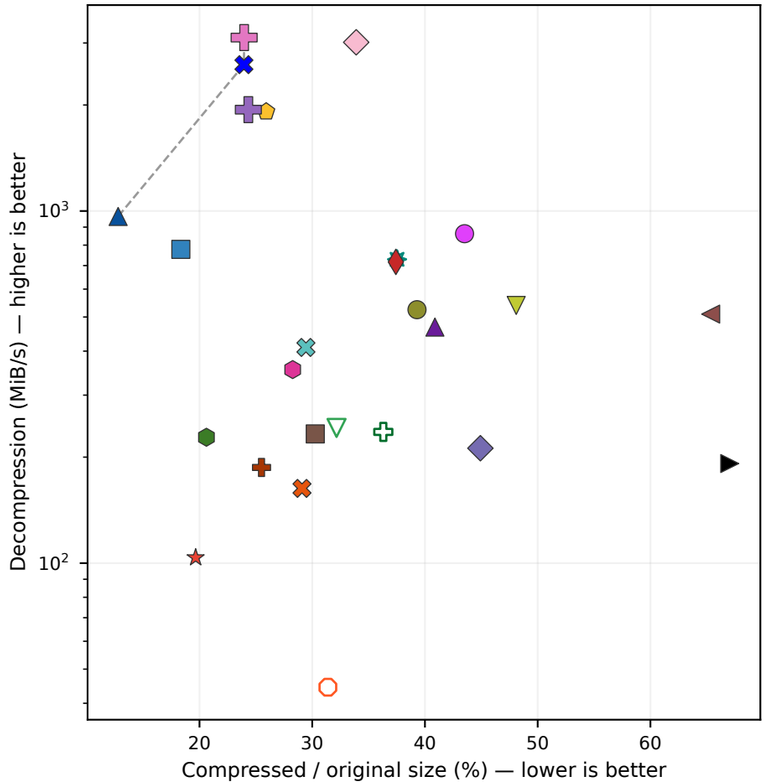
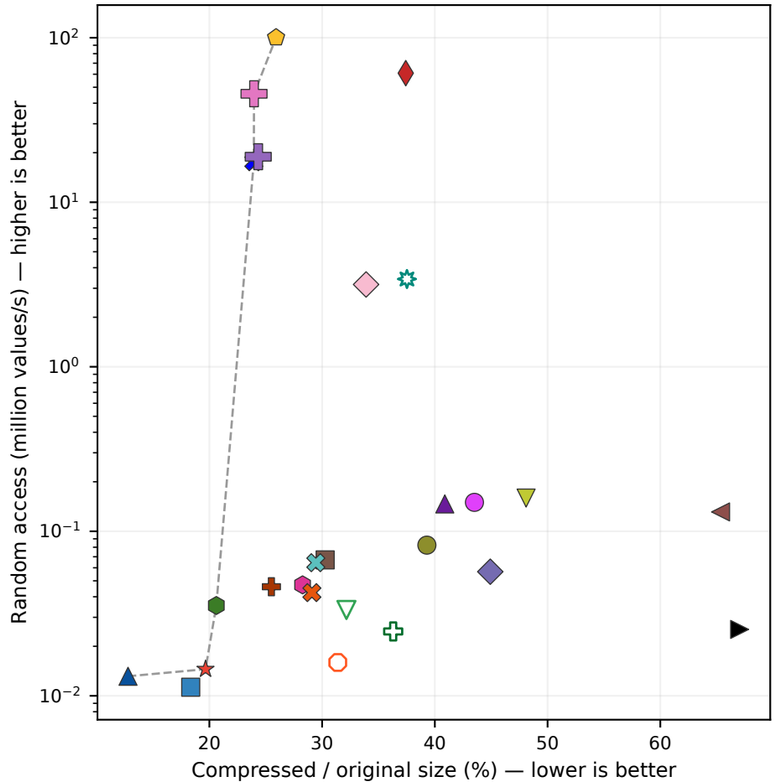



Preliminary: different timing harnesses. ALP/Gorilla/Camel rerun 2026-11-05; original Java Camel, warmed JVM.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Original Camel: not lossless on some inputs. Hollow points excluded; Falcon omitted.

# BM • native GEF 942d3c4 (uniform K=32,000) vs competitors

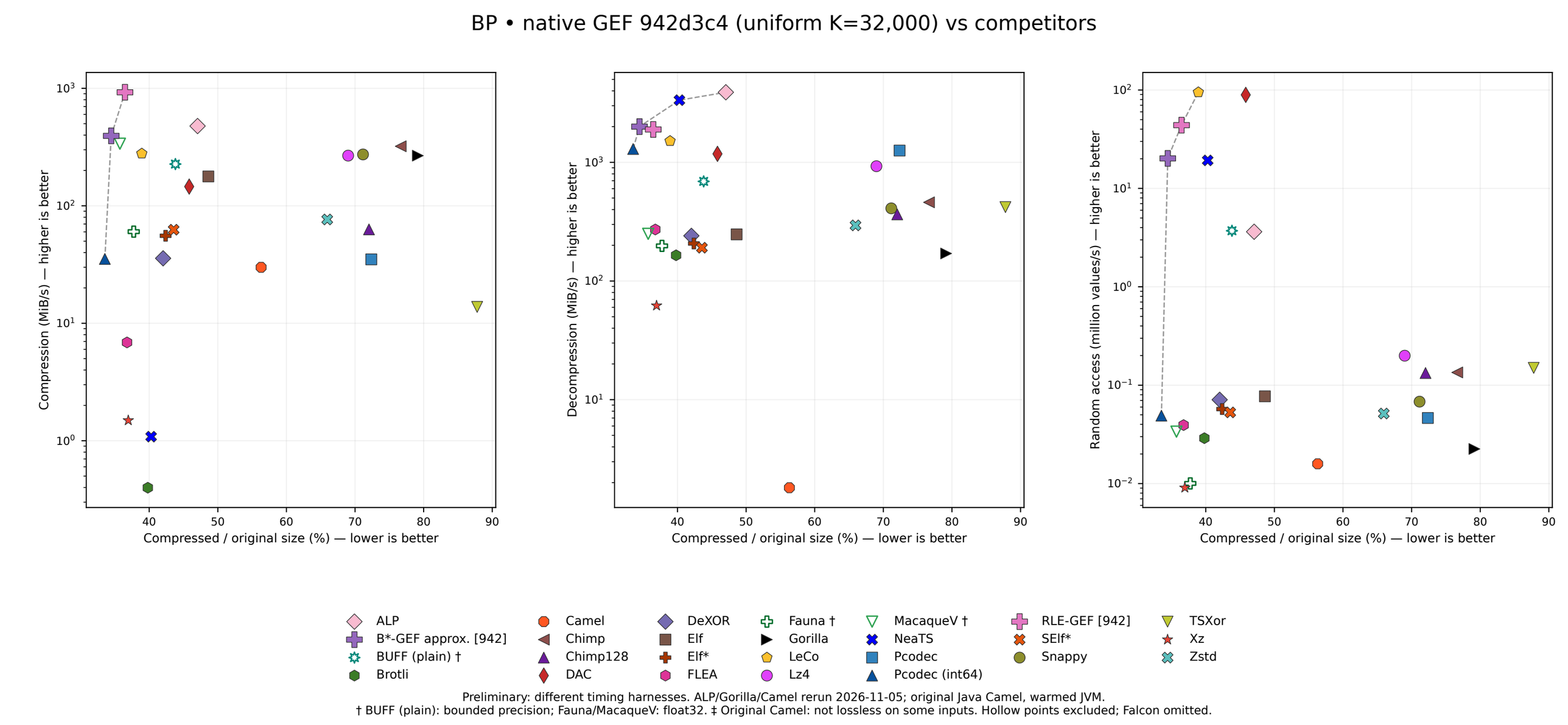


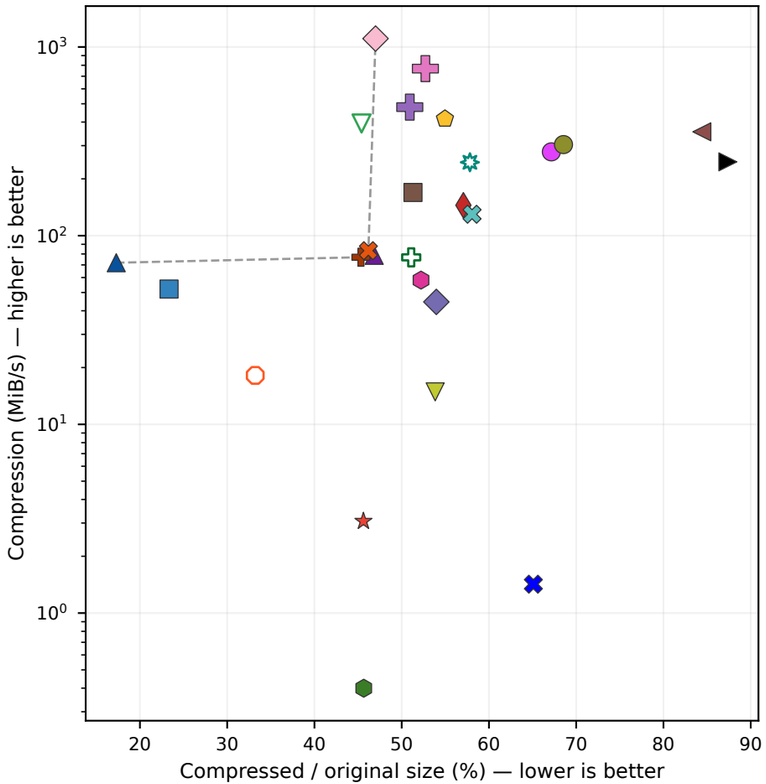



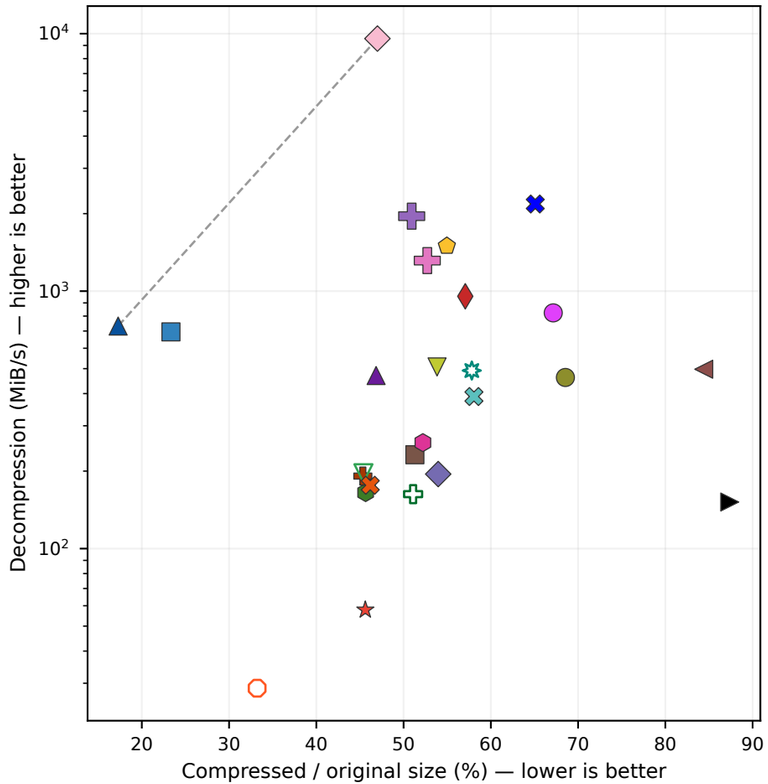

ALP
B*-GEF approx. [942]
BUFF (plain) †
Brotli
Camel ‡
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Fauna †
Gorilla
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [942]
SElf*
Snappy
TSXor
Xz
Zstd

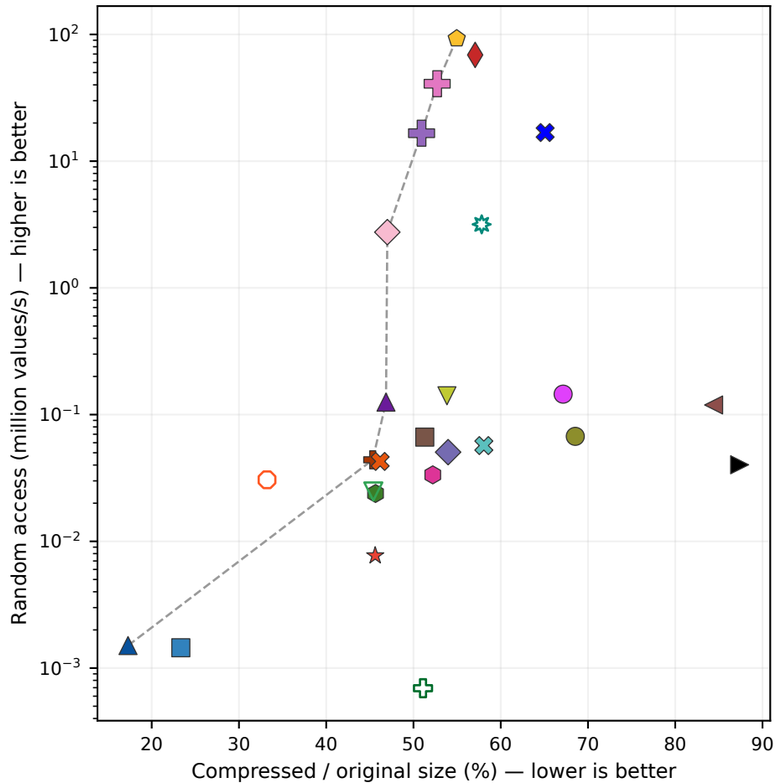
Preliminary: different timing harnesses. ALP/Gorilla/Camel rerun 2026-11-05; original Java Camel, warmed JVM.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Original Camel: not lossless on some inputs. Hollow points excluded; Falcon omitted.


BP • native GEF 942d3c4 (uniform K=32,000) vs competitors
Compression (MiB/s) — higher is better
Decompression (MiB/s) — higher is better
Random access (million values/s) — higher is better
Compressed / original size (%) — lower is better
ALP
B*-GEF approx. [942]
BUFF (plain) †
Brotli
Camel
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Fauna †
Gorilla
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [942]
SElf*
Snappy
TSXor
Xz
Zstd
Preliminary: different timing harnesses. ALP/Gorilla/Camel rerun 2026-11-05; original Java Camel, warmed JVM.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Original Camel: not lossless on some inputs. Hollow points excluded; Falcon omitted.

# BT • native GEF 942d3c4 (uniform K=32,000) vs competitors



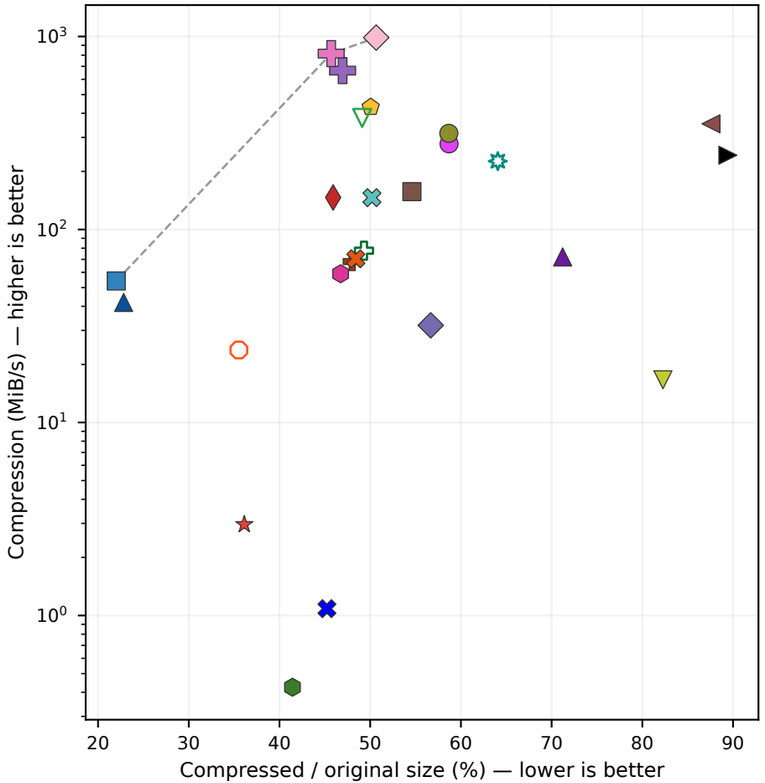



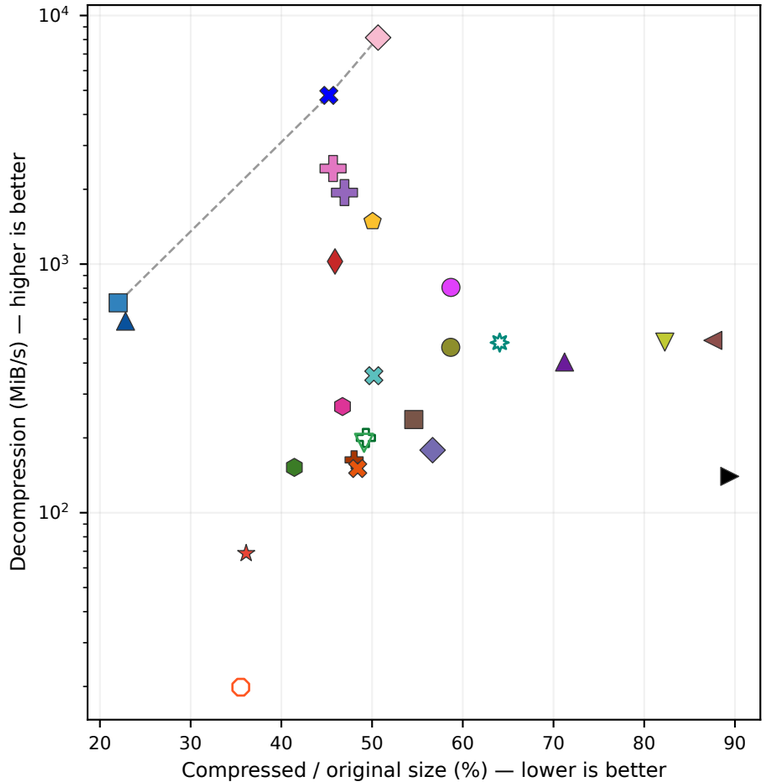



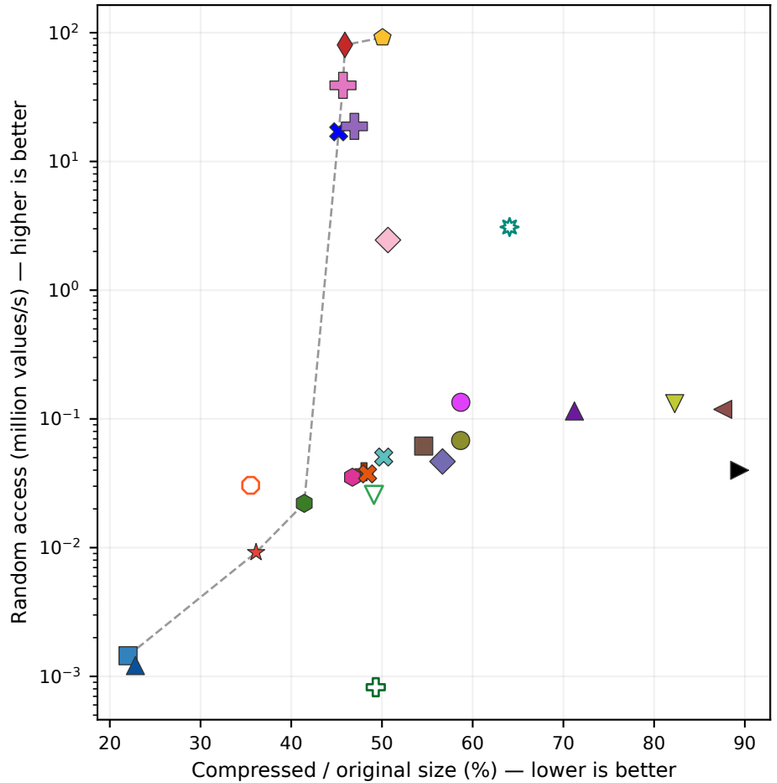

ALP
B*-GEF approx. [942]
BUFF (plain) †
Brotli
Camel ‡
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Fauna †
Gorilla
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [942]
SElf*
Snappy
TSXor
Xz
Zstd

Preliminary: different timing harnesses. ALP/Gorilla/Camel rerun 2026-11-05; original Java Camel, warmed JVM.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Original Camel: not lossless on some inputs. Hollow points excluded; Falcon omitted.

# BW • native GEF 942d3c4 (uniform K=32,000) vs competitors

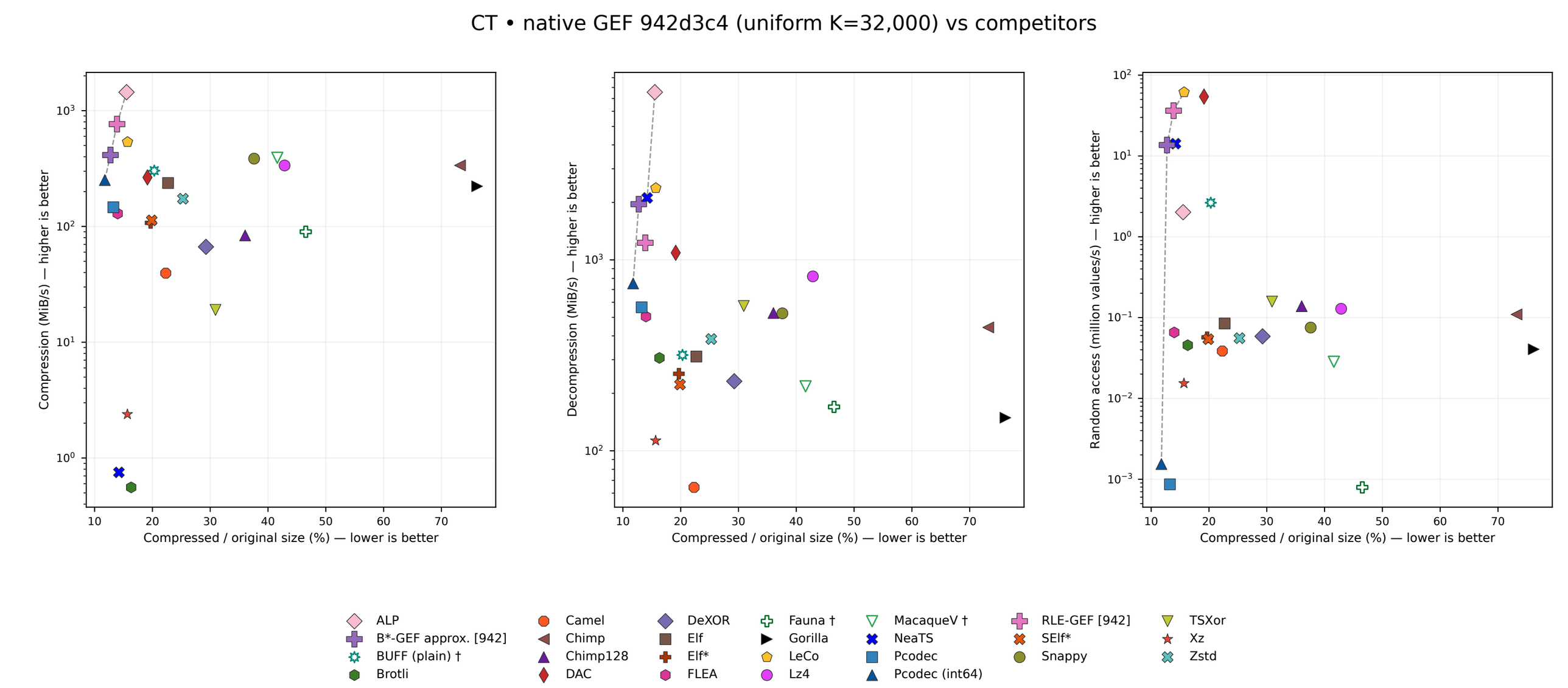


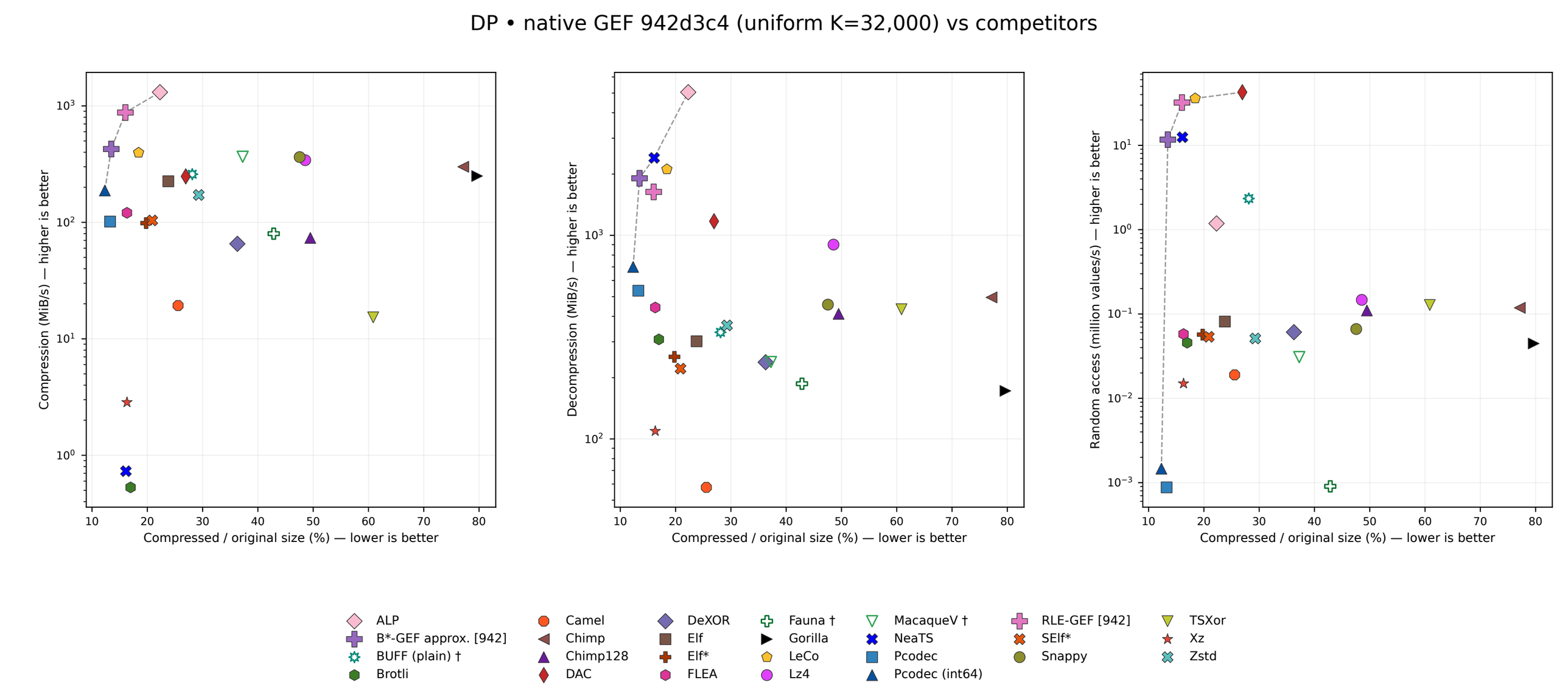



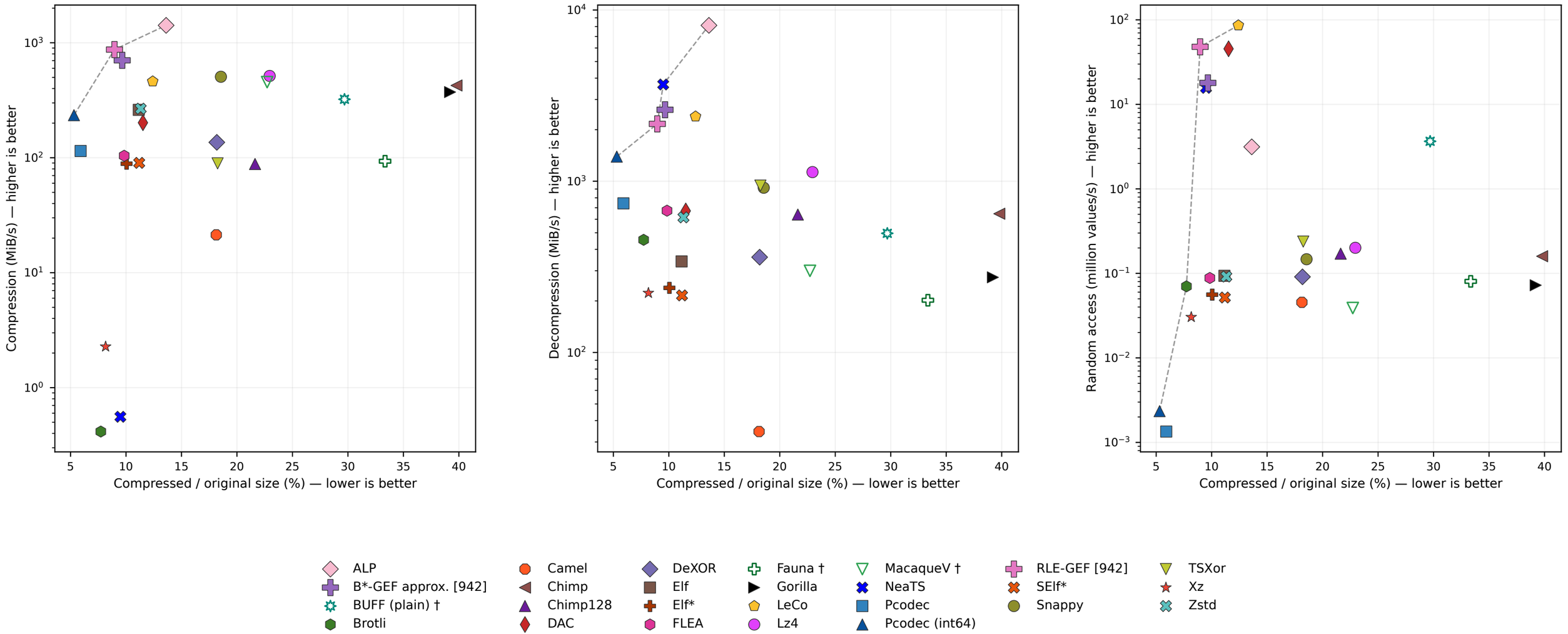




ALP
B*-GEF approx. [942]
BUFF (plain) †
Brotli
Camel ‡
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Fauna †
Gorilla
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [942]
SElf*
Snappy
TSXor
Xz
Zstd

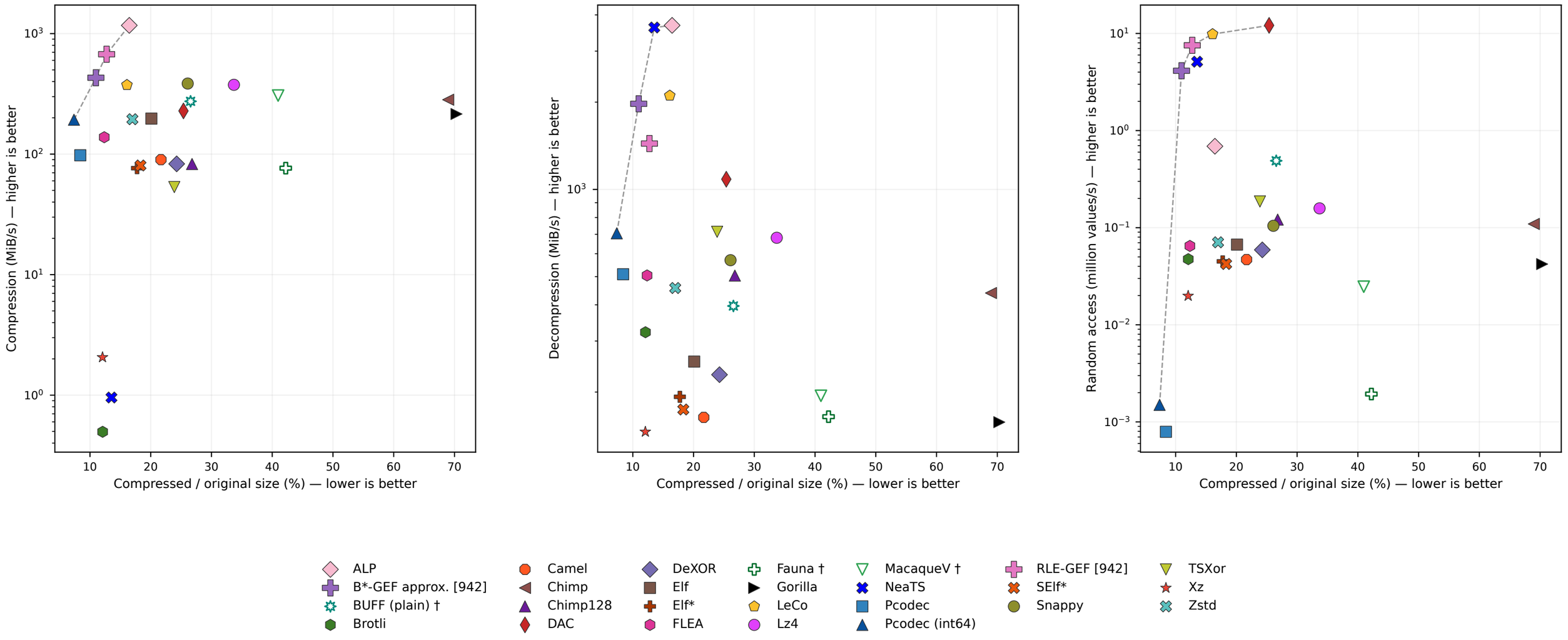
Preliminary: different timing harnesses. ALP/Gorilla/Camel rerun 2026-11-05; original Java Camel, warmed JVM.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Original Camel: not lossless on some inputs. Hollow points excluded; Falcon omitted.


CT • native GEF 942d3c4 (uniform K=32,000) vs competitors
Compression (MiB/s) — higher is better
Decompression (MiB/s) — higher is better
Random access (million values/s) — higher is better
Compressed / original size (%) — lower is better
ALP
B*-GEF approx. [942]
BUFF (plain) †
Brotli
Camel
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Fauna †
Gorilla
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [942]
SElf*
Snappy
TSXor
Xz
Zstd
Preliminary: different timing harnesses. ALP/Gorilla/Camel rerun 2026-11-05; original Java Camel, warmed JVM.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Original Camel: not lossless on some inputs. Hollow points excluded; Falcon omitted.

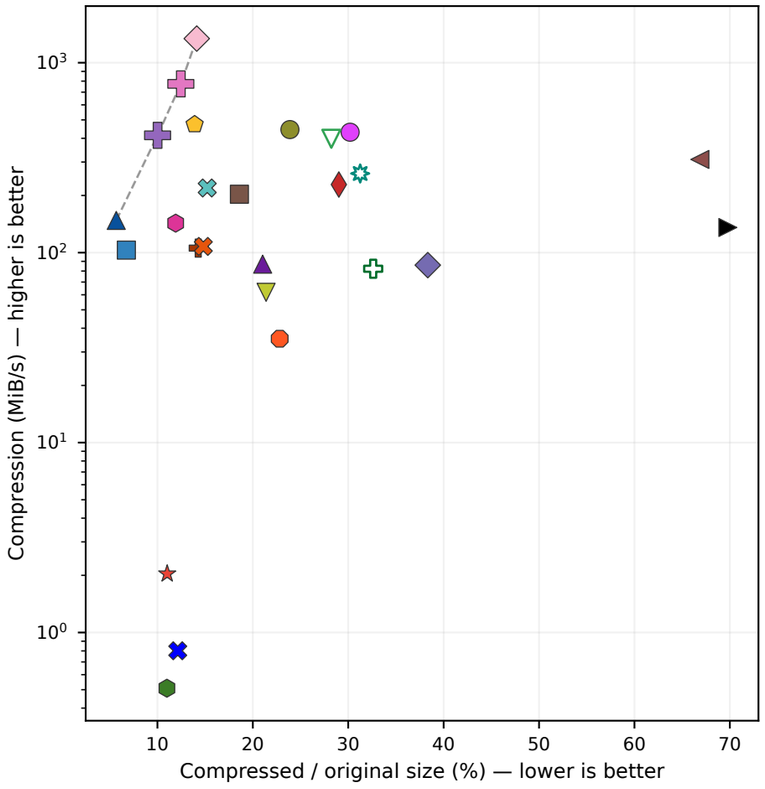

DP • native GEF 942d3c4 (uniform K=32,000) vs competitors
Compression (MiB/s) — higher is better
Decompression (MiB/s) — higher is better
Random access (million values/s) — higher is better
Compressed / original size (%) — lower is better
ALP
B*-GEF approx. [942]
BUFF (plain) †
Brotli
Camel
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Fauna †
Gorilla
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [942]
SElf*
Snappy
TSXor
Xz
Zstd
Preliminary: different timing harnesses. ALP/Gorilla/Camel rerun 2026-11-05; original Java Camel, warmed JVM.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Original Camel: not lossless on some inputs. Hollow points excluded; Falcon omitted.

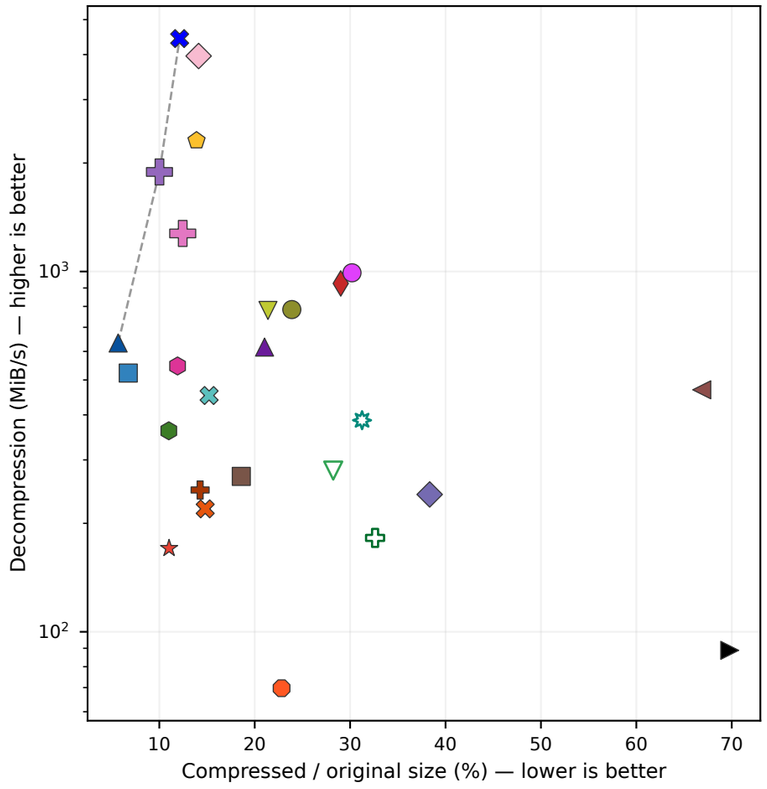


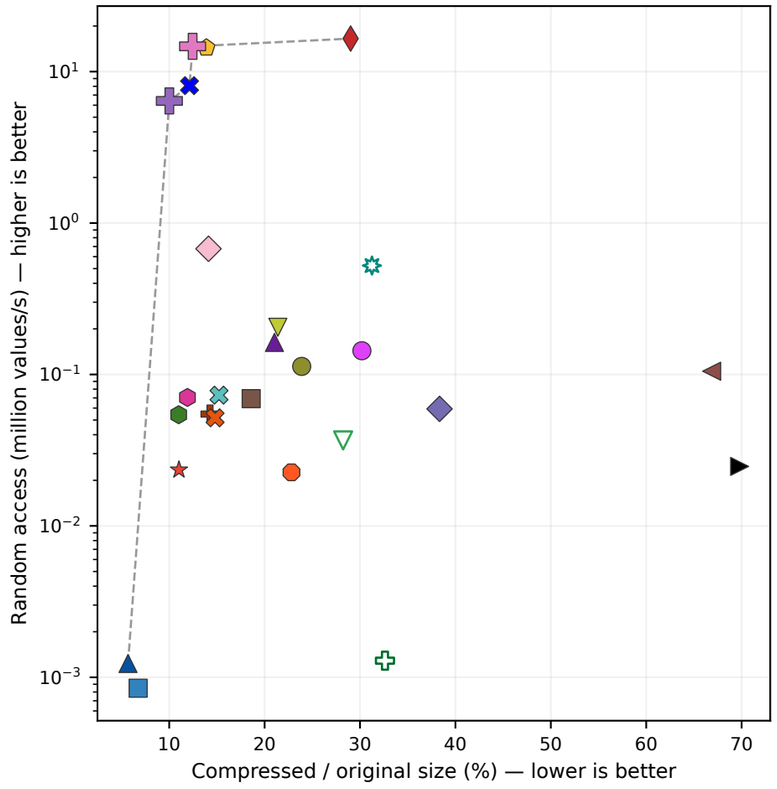

Preliminary: different timing harnesses. ALP/Gorilla/Camel rerun 2026-11-05; original Java Camel, warmed JVM.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Original Camel: not lossless on some inputs. Hollow points excluded; Falcon omitted.


ECG • native GEF 942d3c4 (uniform K=32,000) vs competitors
Compression (MiB/s) — higher is better
Decompression (MiB/s) — higher is better
Random access (million values/s) — higher is better
Compressed / original size (%) — lower is better
ALP
B*-GEF approx. [942]
BUFF (plain) †
Brotli
Camel
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Fauna †
Gorilla
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [942]
SElf*
Snappy
TSXor
Xz
Zstd
Preliminary: different timing harnesses. ALP/Gorilla/Camel rerun 2026-11-05; original Java Camel, warmed JVM.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Original Camel: not lossless on some inputs. Hollow points excluded; Falcon omitted.

# GE • native GEF 942d3c4 (uniform K=32,000) vs competitors



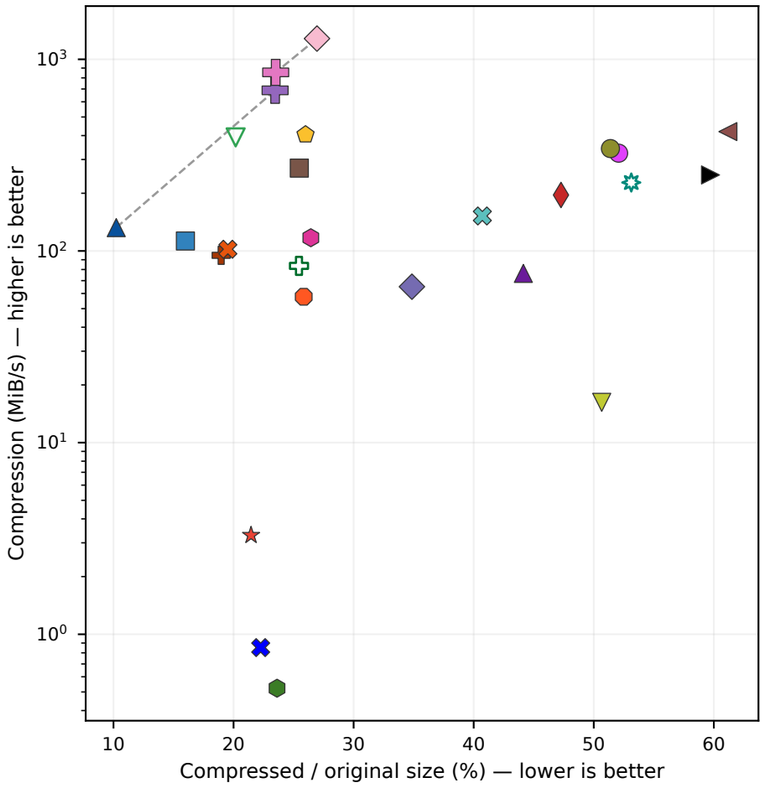



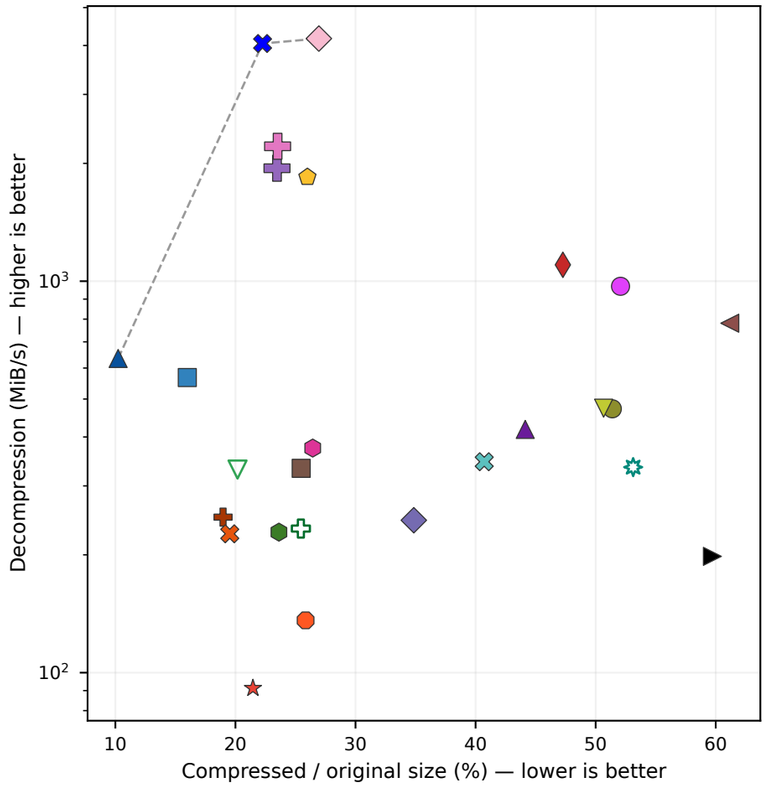



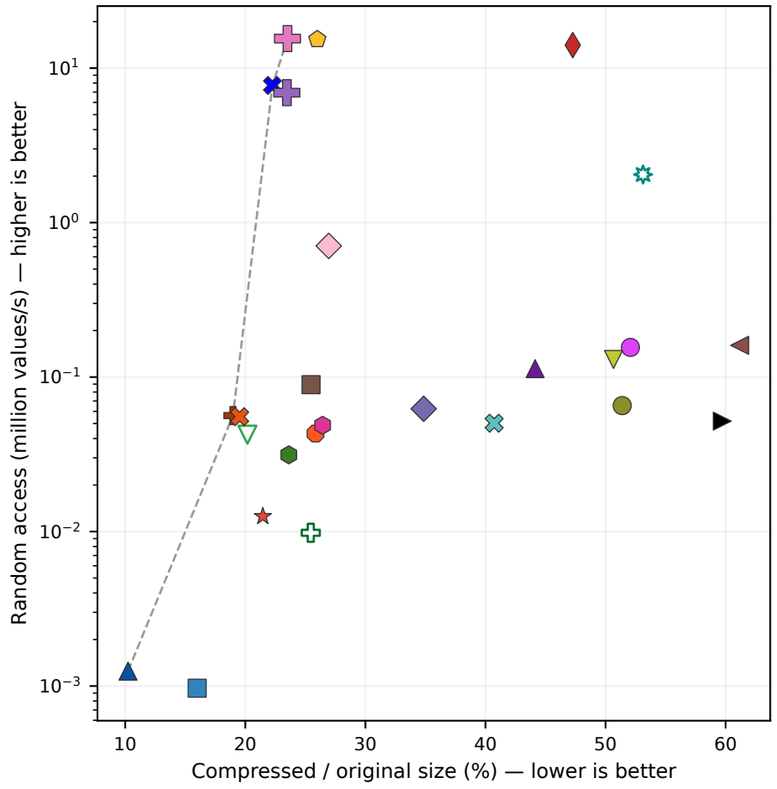

ALP
B*-GEF approx. [942]
BUFF (plain) †
Brotli
Camel
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Fauna †
Gorilla
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [942]
SElf*
Snappy
TSXor
Xz
Zstd

Preliminary: different timing harnesses. ALP/Gorilla/Camel rerun 2026-11-05; original Java Camel, warmed JVM.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Original Camel: not lossless on some inputs. Hollow points excluded; Falcon omitted.

# LAT • native GEF 942d3c4 (uniform K=32,000) vs competitors



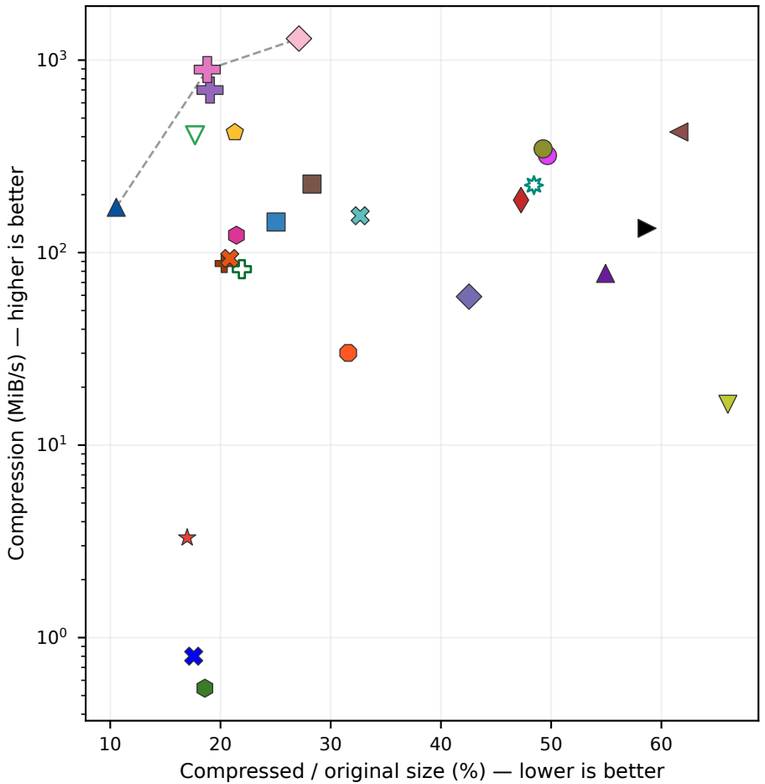



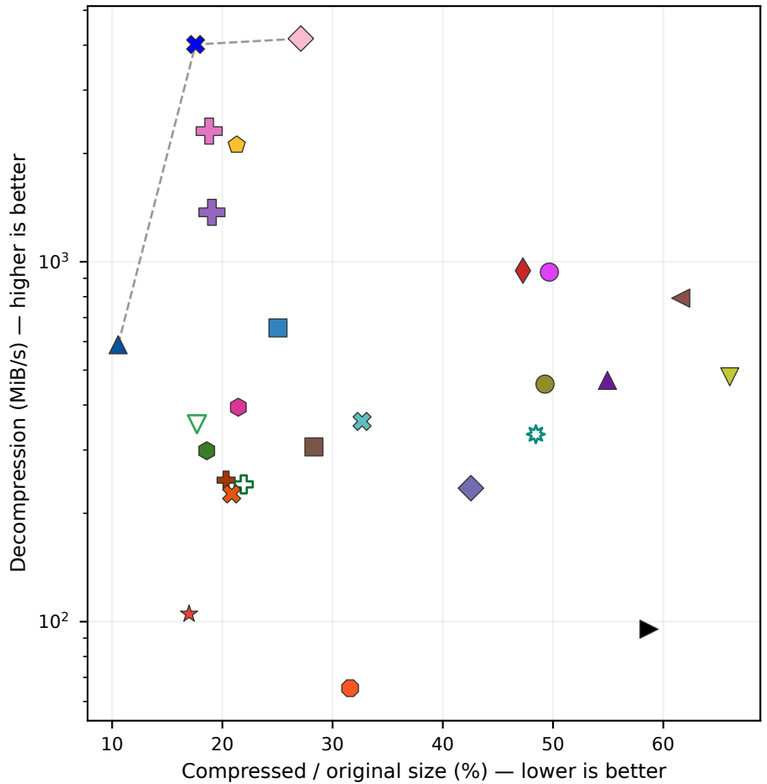



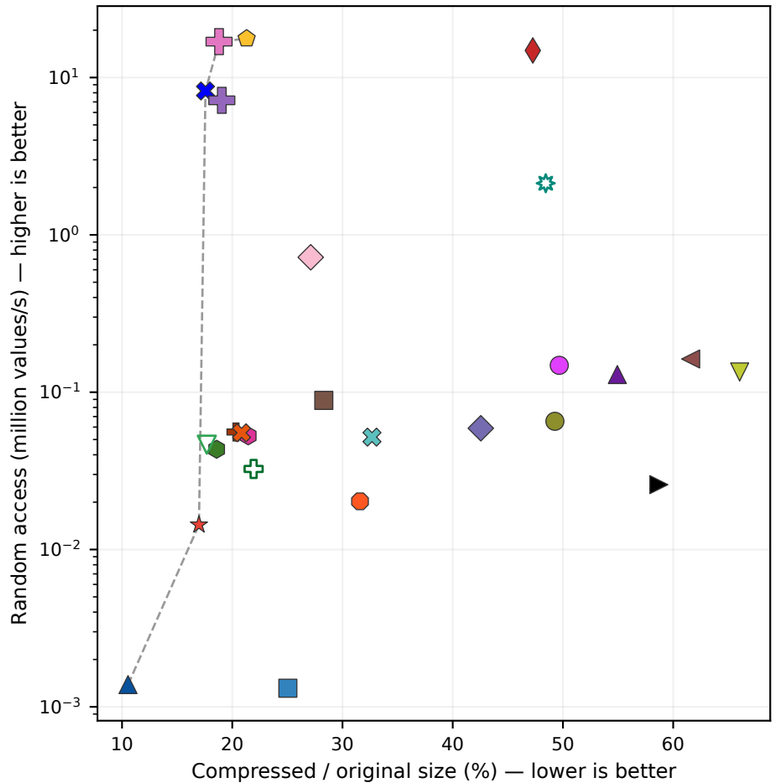

ALP
B*-GEF approx. [942]
BUFF (plain) †
Brotli
Camel
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Fauna †
Gorilla
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [942]
SElf*
Snappy
TSXor
Xz
Zstd

Preliminary: different timing harnesses. ALP/Gorilla/Camel rerun 2026-11-05; original Java Camel, warmed JVM.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Original Camel: not lossless on some inputs. Hollow points excluded; Falcon omitted.

# LON • native GEF 942d3c4 (uniform K=32,000) vs competitors

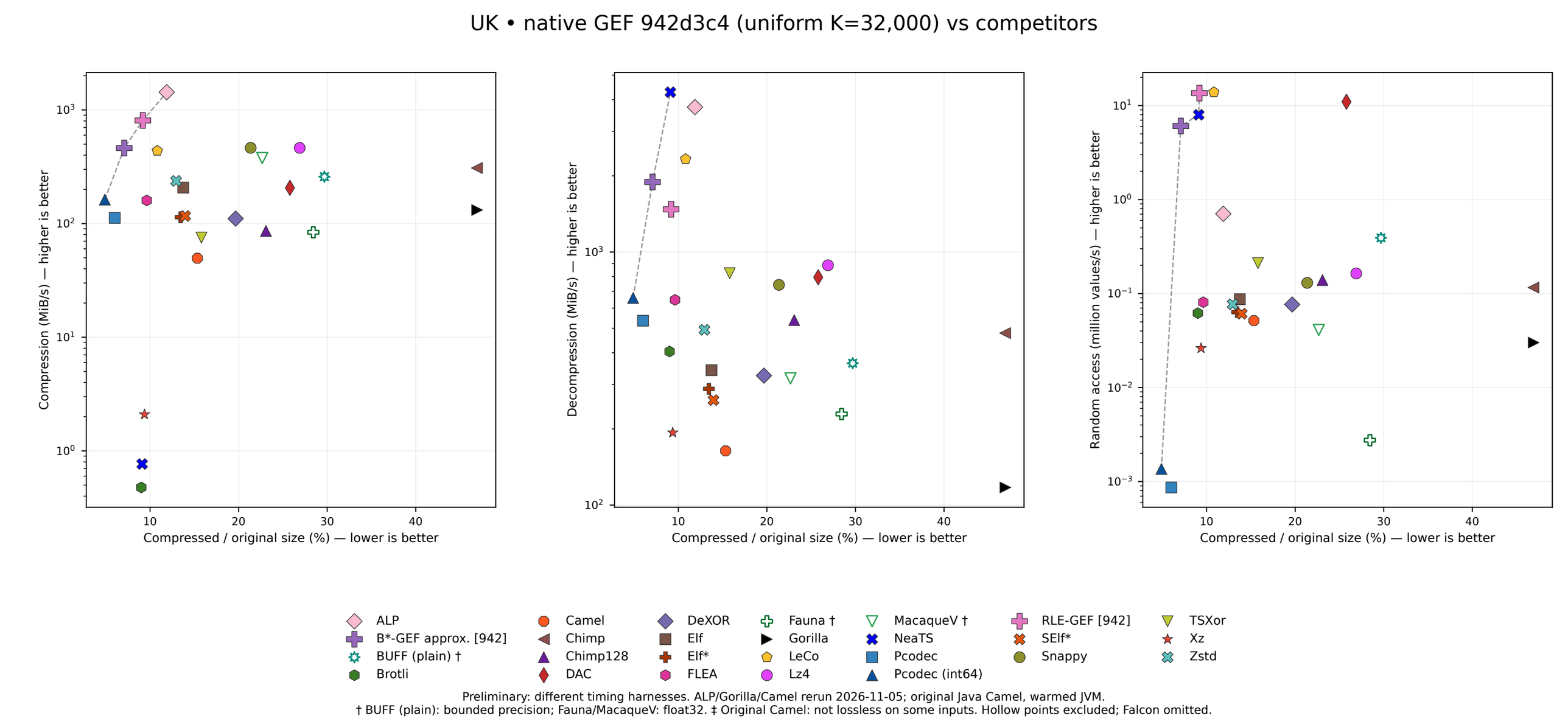


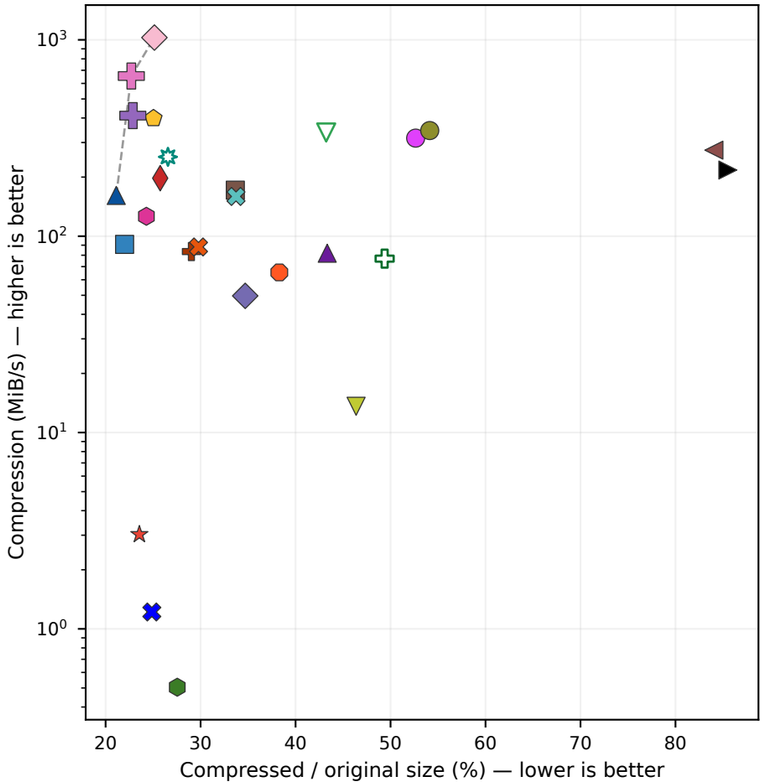



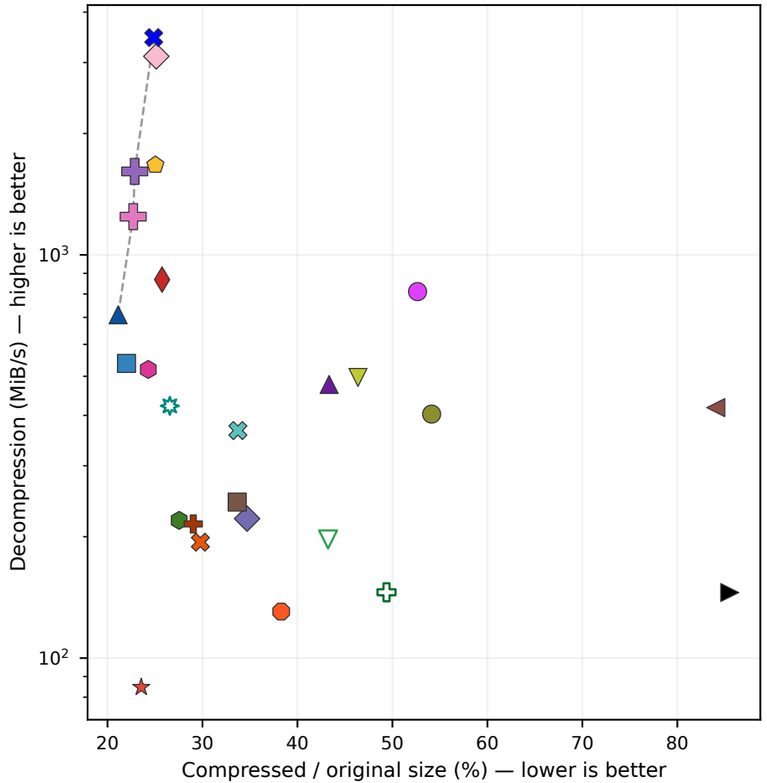

ALP
B*-GEF approx. [942]
BUFF (plain) †
Brotli
Camel
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Fauna †
Gorilla
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [942]
SElf*
Snappy
TSXor
Xz
Zstd

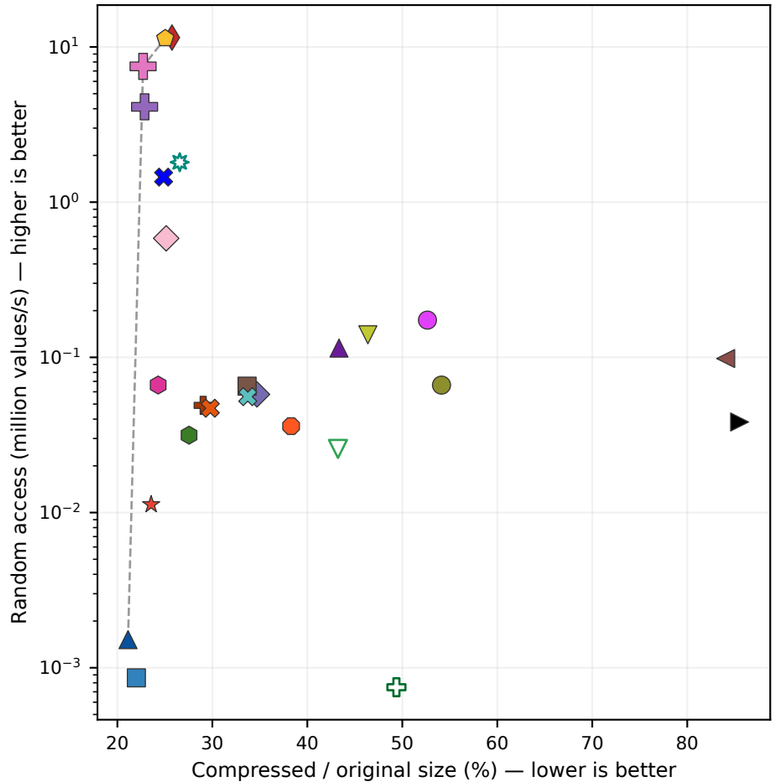
Preliminary: different timing harnesses. ALP/Gorilla/Camel rerun 2026-11-05; original Java Camel, warmed JVM.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Original Camel: not lossless on some inputs. Hollow points excluded; Falcon omitted.


UK • native GEF 942d3c4 (uniform K=32,000) vs competitors
Compression (MiB/s) — higher is better
Decompression (MiB/s) — higher is better
Random access (million values/s) — higher is better
Compressed / original size (%) — lower is better
ALP
B*-GEF approx. [942]
BUFF (plain) †
Brotli
Camel
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Fauna †
Gorilla
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [942]
SElf*
Snappy
TSXor
Xz
Zstd
Preliminary: different timing harnesses. ALP/Gorilla/Camel rerun 2026-11-05; original Java Camel, warmed JVM.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Original Camel: not lossless on some inputs. Hollow points excluded; Falcon omitted.

# WD • native GEF 942d3c4 (uniform K=32,000) vs competitors




ALP
B*-GEF approx. [942]
BUFF (plain) †
Brotli
Camel
Chimp
Chimp128
DAC
DeXOR
Elf
Elf*
FLEA
Fauna †
Gorilla
LeCo
Lz4
MacaqueV †
NeaTS
Pcodec
Pcodec (int64)
RLE-GEF [942]
SElf*
Snappy
TSXor
Xz
Zstd

Preliminary: different timing harnesses. ALP/Gorilla/Camel rerun 2026-11-05; original Java Camel, warmed JVM.
† BUFF (plain): bounded precision; Fauna/MacaqueV: float32. ‡ Original Camel: not lossless on some inputs. Hollow points excluded; Falcon omitted.